

Mridulla Ganesh

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EDUCATION

Purdue University

Computer Engineering

West Lafayette, IN

August 2023 — May 2027

- Cumulative GPA: 3.4/4.0 | Dean's List, Semester Honors
- Relevant Coursework: ASIC Design, Microprocessor Systems, Data Structures, Computer Architecture

WORK EXPERIENCE

New Product Development Co-Op

May 2026 — Present

Advanced Micro Devices

Austin, TX

- Extended firmware emulation code to model thermal and power behavior of AMD desktop processors, enabling pre-silicon power and thermal predictions
- Integrated emulation infrastructure with System Level Test workflows to predict real-world behavior
- Supported SLT bring-up and validation through debugging, characterization, and system-level test development

Teaching Assistant, Embedded Systems Lab

August 2025 — May 2026

Purdue University

West Lafayette, IN

- Mentored students in embedded C programming for RP2350 microcontrollers, teaching real-time concepts, interrupt handling, and register-level hardware interfacing
- Debugged hardware-software integration issues across GPIO, UART, SPI, I2C, timers, and interrupts using oscilloscopes and JTAG debuggers
- Guided students in root-cause analysis of timing issues, memory-related failures, and protocol-level communication problems

Quality Engineering Intern

May 2025 — August 2025

Coherent, Inc.

Monroe, CT

- Developed Python/OpenCV pipelines to process image data for AI-based defect detection of semiconductor wafer chucks
- Performed hardware failure analysis through physical inspection and root cause investigation of component defects

RESEARCH EXPERIENCE

Undergraduate Research Assistant, AI Network Performance

January 2025 — Present

Purdue University, w/ Dr. Xiaoqi Chen

West Lafayette, IN

- Evaluated SimAI for simulating NCCL collective communication in distributed AI training, identifying accuracy gaps in collective operation modeling
- Designed and executed experiments on our GPU cluster using ns-3 and Astra-Sim; analyzed results using perf, Flamegraph, and TensorBoard to characterize bottlenecks and validate findings for peer-reviewed publications
- Architected and implemented an MSCCL++-based extension to SimAI to improve collective communication simulation accuracy for distributed training workloads

Research with VIP SoCET, Datacenter Network Accelerator

August 2025 — Present

Purdue University

West Lafayette, IN

- Developed a Python-based emulation framework for a custom SmartNIC to enable pre-silicon software development and validation
- Implemented Ethernet and RDMA communication layers to bridge host software with accelerator hardware, enabling hardware-software co-design and system-level testing
- Designed workload-driven benchmarks using AICB to characterize SmartNIC functionality, packet processing behavior, and RDMA offload performance

PROJECTS

Multicore RISC-V Processor with Cache Coherence

January 2026 — May 2026

- Designed and implemented a multicore 5-stage pipelined RISC-V processor in SystemVerilog with instruction and data cache hierarchies
- Developed cache coherence protocols to maintain memory consistency across processor cores
- Implemented pipeline control logic including hazard detection, forwarding, and branch handling
- Built simulation-based verification infrastructure to validate processor functionality and architectural performance

SKILLS

- **Programming Languages:** C, C++, Python, CUDA, Java, MATLAB, RISC-V Assembly
- **Hardware Design:** SystemVerilog, RTL Design, Digital Logic Design, Computer Architecture
- **Networking:** NCCL, RDMA, ns-3, htsim, Astra-Sim, Collective Communication, Network Emulation/Simulation
- **Tools:** Linux, Git, LTspice, Autodesk Eagle